Replication of Card and Krueger (1994), Minimum Wages and Employment in New Jersey and Pennsylvania

1. Research question and motivation

On 1 April 1992 New Jersey raised its minimum wage from 4.25 to 5.05 dollars an hour, while neighbouring eastern Pennsylvania kept its rate at 4.25 dollars. The two areas share the same labour market along the Delaware River, which makes the increase a natural experiment. Card and Krueger ask whether it reduced employment in the fast-food industry, and their published answer is no: employment in New Jersey restaurants rose slightly relative to Pennsylvania. This project reproduces that result from the raw survey data, compares every reproduced value with the published value, and adds one extension.

2. Data sources, sample construction, and variable definitions

The analysis uses one store-level survey file, public.csv, distributed with the course and publicly available from the original authors. It holds 410 restaurants and 46 fields: 331 in New Jersey and 79 in eastern Pennsylvania, across four chains, each surveyed twice by telephone, in February and March 1992 and again in November and December 1992. In the second wave 399 stores responded, six had closed permanently, five were temporarily closed, and one refused.

Employment is measured on a full-time-equivalent basis: full-time employees plus managers and assistant managers, with each part-time employee counted as half a position. Other variables come straight from the survey, including the starting wage, weekday hours open, cash registers, registers open at eleven o'clock, recruiting bonus and meal programs. The full-meal price adds a medium soda, a small fries and an entree, all including tax, and the share of full-time workers is full-time employees divided by full-time-equivalent employment. The wage gap is the proportional increase in a New Jersey store's starting wage needed to reach 5.05 dollars, and is zero for Pennsylvania stores.

Permanently closed stores are given zero employment in the second wave and temporarily closed stores are left missing, as in the paper. The regression sample has 357 stores and the balanced sample used in the difference-in-differences table has 384, both matching the paper.

3. Empirical design

The design is a difference-in-differences comparison: the change in average employment in New Jersey minus the change in Pennsylvania. The paper also writes the same comparison as a regression of the employment change on either a New Jersey indicator or the wage gap, so that the coefficient reads as the employment change associated with a one-unit change in the treatment variable. Four specifications are used, with the indicator

or the gap, alone or with controls for chain and ownership, and a fifth adds four regional indicators, two for northern and central New Jersey and two for the Pennsylvania areas.

A set of robustness checks re-estimates the same regressions under alternative definitions, varying the weight on part-time workers, dropping managers, dropping New Jersey shore stores or stores needing more than two calls, using the proportional rather than the absolute change, weighting by initial employment, and using Pennsylvania stores only with a hypothetical gap. The same structure is applied to other outcomes and to the full-meal price.

4. Core replication results and comparison with the original paper

The replication matches the paper closely. Each table below puts the replicated value next to the value printed in the paper, and the paragraphs between the tables read them. Coefficients are reported with their standard errors in brackets.

Starting wages and employment (the paper's Tables 2 and 3)

Item	Replication	Published
New Jersey starting wage, before	4.61	4.61
New Jersey starting wage, after	5.08	5.08
Pennsylvania starting wage, before	4.63	4.63
Pennsylvania starting wage, after	4.62	4.62
New Jersey employment, before	20.44	20.4
New Jersey employment, after	21.03	21.0
Pennsylvania employment, before	23.33	23.3
Pennsylvania employment, after	21.17	21.2
Change in New Jersey employment	0.59	0.59
Change in Pennsylvania employment	-2.17	-2.16
Raw difference-in-differences	2.75	2.76
Difference-in-differences, balanced sample	2.75	2.75
Difference-in-differences, temporarily closed stores set to zero	2.51	2.51

New Jersey's average starting wage rose from 4.61 to 5.08 dollars while Pennsylvania's barely moved, so the increase did bite. Employment in New Jersey edged up by 0.59 full-time equivalents while Pennsylvania fell by 2.17, which gives a raw difference-in-differences of 2.75. The estimate is unchanged on the balanced sample and falls only to 2.51 once temporarily closed stores are counted as having zero employment. On the evidence of the first table, employment in New Jersey did not fall.

Reduced-form employment regressions (the paper's Table 4)

Specification	Replication	Published
New Jersey indicator alone	2.326 (1.192)	2.33 (1.19)

New Jersey indicator with controls	2.304 (1.196)	2.30 (1.20)
Wage gap alone	15.653 (6.080)	15.65 (6.08)
Wage gap with controls	14.916 (6.205)	14.92 (6.21)
Wage gap with controls and regions	11.979 (7.419)	11.91 (7.39)
Sample size	357	357
Mean of the change in employment	-0.238	-0.237
Standard deviation of the change in employment	8.825	8.825

Both ways of writing the treatment give the same answer. The New Jersey indicator is about 2.3 with or without controls, and the wage gap implies an employment change of twelve to sixteen full-time equivalents for a one-unit change in the gap. The estimates rest on the same 357 stores, whose employment change averages negative 0.238 with a standard deviation of 8.825, so the standard errors are large relative to the coefficients. The estimates are therefore best read as ruling out a substantial employment decline rather than as precise measurements of a gain.

Robustness checks (the paper's Table 5)

Specification	**Replication**	**Published**
Temporarily closed stores set to zero, New Jersey indicator	2.028 (1.219)	2.20 (1.21)
Temporarily closed stores set to zero, wage gap	14.424 (6.312)	14.42 (6.31)
Managers excluded, New Jersey indicator	2.342 (1.166)	2.27 (1.17)
Part-time workers at four tenths of a position, New Jersey indicator	2.340 (1.200)	2.34 (1.20)
Part-time workers at six tenths of a position, New Jersey indicator	2.268 (1.207)	2.27 (1.21)
New Jersey shore stores excluded, New Jersey indicator	2.585 (1.195)	2.58 (1.19)
Stores needing more than two calls excluded, New Jersey indicator	2.419 (1.286)	2.41 (1.28)
Pennsylvania stores only, wage gap	-0.304 (22.003)	-0.30 (22.00)

Every alternative definition leaves the New Jersey indicator positive and roughly between 2.0 and 2.6, so the conclusion does not depend on how part-time workers are weighted, on whether managers are counted, or on which stores are kept in the sample. One cell does not match. The New Jersey indicator in the row that sets temporarily closed stores to zero is 2.028 here against 2.20 in the paper. The wage-gap coefficient in that same row reproduces at 14.424 against a published 14.420, the standard error at 1.219 against 1.21, and the other New Jersey values in the table all match, so the printed cell does not look consistent with the rest of its own row. It is reported as it stands rather than adjusted to look closer to the paper.

Other outcomes and prices (the paper's Tables 6 and 7)

Outcome, New Jersey minus Pennsylvania	Replication	Published
Share of full-time workers, percentage points	7.30	7.29
Hours open per weekday	-0.11	-0.11
Number of cash registers	-0.17	-0.17
Registers open at eleven o'clock	0.18	0.17
Low-price meal program	-3.39	-3.39
Free meal program	2.00	2.00
Combination of meals	1.08	1.09
Time to the first pay increase	2.51	2.51
Size of the first pay increase	0.01	0.01
Slope of the wage profile	0.00	-0.01

Specification, full-meal price	Replication	Published
New Jersey indicator alone	0.0325 (0.0144)	0.033 (0.014)
New Jersey indicator with controls	0.0367 (0.0140)	0.037 (0.014)
Wage gap alone	0.0778 (0.0748)	0.077 (0.075)
Wage gap with controls	0.1470 (0.0740)	0.146 (0.074)
Wage gap with controls and regions	0.0659 (0.0893)	0.063 (0.089)
Sample size	317	315

The two states differ by 7.30 percentage points in the change in the share of full-time workers, and the remaining rows reproduce the published figures, including the small rise in the number of registers open at eleven o'clock. Prices rise in every specification, by roughly three per cent when the New Jersey indicator is used and by more when the wage gap is used, on a sample of 317 stores against 315 in the paper; the paper reads the same pattern as a small pass-through of the higher wage into prices.

Taken together, 91 of the 92 compared values fall inside the tolerance allowed for the rounding used in the paper, and the single exception is the cell described above. Five items cannot be reproduced with this data: the sampling frame of Table 1, the Newark and Camden zip-code subsamples, the McDonald's entry analysis, the state employment-population evidence from the Current Population Survey, and the cash-register instrumental-variables check.

5. Independent extension

The extension is a placebo exercise for the main employment estimate. It asks how large a coefficient the same regression would typically produce if the state labels carried no information. To find out, the New Jersey and Pennsylvania labels are reassigned at random inside the estimation sample 1,000 times, holding the number of labelled stores fixed, and the reduced-form regression with controls is re-estimated each time. The random seed is fixed, so the exercise is reproducible.

The observed coefficient is 2.304, while the placebo estimates average 0.015 with a standard deviation of 1.196. Only 28 of the 1,000 placebo estimates reach or exceed the observed value, which is a one-sided p-value of 0.028, and the range covering 95 per cent of the placebo estimates runs from negative 2.182 to 2.403. The result supports the paper's conclusion that the relative employment gain in New Jersey reflects the change in the minimum wage rather than chance differences between the two groups of stores. It is a descriptive exercise rather than exact inference, because the stores were not in fact assigned to states at random.

6. Validation, limitations, and conclusion

The replication was validated in three ways. Every value is compared with the published one automatically and the comparison is written to a file, so it can be checked without reading the code. The script was run repeatedly, from a clean output state and from different directory locations, and produced identical results. And the published benchmarks were read twice and cross-checked wherever the paper reports the same quantity in two places, which is how the single flagged cell was identified.

Three limitations are worth stating. Parts of the paper that need data outside the survey file are not reconstructed. The survey carries the usual problems of a telephone survey, including item non-response and a second wave collected seven to eight months after the increase. And the extension is a placebo exercise rather than a formal test, resting on the assumption that the two groups of stores were comparable before the increase, which the first-wave means support.

The overall conclusion is that the paper's results hold up. The minimum wage increase in New Jersey did not reduce employment in the fast-food industry: relative to the Pennsylvania restaurants, New Jersey ended the period with slightly more full-time-equivalent employment, not less, and the estimated effect is positive in every specification that was run.